

RAGHAV KAUL

(602) 587-6935 • kaulraghav@gmail.com • [linkedin.com/in/raghavkaul](https://www.linkedin.com/in/raghavkaul) • [kaulraghav.github.io/](https://github.com/raghavkaul)

SUMMARY

Software engineer with 4+ years of experience building and operating large-scale, cloud-based distributed systems serving real-time traffic. Team-player focused on backend and middleware development, shipping customer-facing features end-to-end, designing data pipelines, leading security testing initiatives, and reducing operational costs.

EXPERIENCE

Amazon Web Services

Software Engineer II, Mountain View, California

Apr. 2025– Present

- Owned a campaign to retire a [legacy service](#), eliminating test resources across 600+ accounts and achieving annual cost savings of \$60K; developed custom **Python** scripts to enable scalable future cleanups.
- Redesigned media streaming infrastructure in **TypeScript** and **Node.js** to support mid-call session updates, unblocking an enterprise customer's real-time call recording compliance across all production regions.
- Led a two-engineer team to build an automated security testing framework in **Java**, enabling comprehensive authentication and authorization validation across **Amazon Connect** telephony production APIs; delivered an org-wide presentation on development best practices for broader team adoption.

Software Engineer I, Santa Clara, California

Sep. 2022 – Apr. 2025

- Analyzed monitoring usage patterns for expensive canaries and tests, reducing costs by 60–95% across all production-regions using **Java** and **AWS CloudFormation** while maintaining full visibility and operational safety.
- Worked in a two-person team to implement security auditing canaries for 60 **Amazon Chime SDK** voice [APIs](#) using **Java**, safely updating service logic to meet strict scheduling requirements and ensure company-wide compliance.
- Designed an automated integration testing system in **Java** and **Python** with intelligent retry logic and failure detection, reducing manual testing effort by 16 hours per month and enabling reliable continuous deployment.

Data Engineer, Rhombus Power Inc., Mountain View, California

Oct. 2021 – Sep. 2022

- Built automated **AWS** data pipelines with Python to ingest 100K+ records from diverse government and trade sources, standardizing and consolidating data into a centralized **PostgreSQL** database.
- Developed an automated scoring system using **cron jobs** to assess operational readiness of multiple international airports based on categorized regulatory notices and site-specific constraints.
- Built a time-lagged correlation dashboard with **pandas** and **seaborn** to identify key cross-market trends, enabling data-driven decisions for executive leadership.
- Authored web scrapers using the **Twitter API** and **Python** ETL pipelines to automate data extraction, transformation, and pipeline integration into a centralized database.

TECHNICAL SKILLS

Languages: Proficient: Python, JAVA (Spring), TypeScript (Node.js), C++, SQL, Bash; Familiar: JavaScript, HTML, CSS

Tools/Frameworks: AWS (Lambda, EC2, DynamoDB, CloudWatch), Git, Linux, Tableau, POSTMAN, Agentic AI, WordPress

PROJECT EXPERIENCE

Meal Detection using CGM data (Python, NumPy, Pandas, SciPy, scikit-learn, pickle) ([link](#))

Jan. 2020– May 2020

- Built supervised and unsupervised ML models on continuous glucose monitor data to detect meal intake in diabetic patients, achieving 81% accuracy with linear SVC.

Compiler Design Projects (C++, Bash Scripting, Ubuntu)

Sep. 2019 – Dec. 2019

- Implemented a compiler from scratch including parser, type checker, and instruction node linker for I/O, assignment, and looping constructs.

EDUCATION

Arizona State University, Tempe, Arizona

Aug 2019 – May 2021

Master of Science in Computer Science, **GPA: 3.75/4**

Activities: Graduate TA, CSE 340 under Dr. Fainekos, Spring 2022

Manipal University Jaipur, Rajasthan, India

Aug 2015 - May 2019

Bachelor of Technology in Computer Science, **GPA: 8.14/10**

Activities: President, The Music Club 2017-2018; Class Representative 2015-2017